Local Decoder.

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Introduction

Today:

- Proof.
  1. Local Decoders.

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$-depolarized channel:

- There exists function $M:(0,1)\to(0,1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n,k)-Toric, then both the swift-rule and the flip upon majority are local decoders.

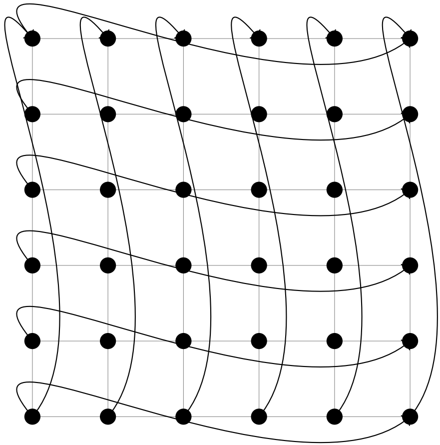


Figure: On the left is the Toric Graph. On the right are cross and face checks.

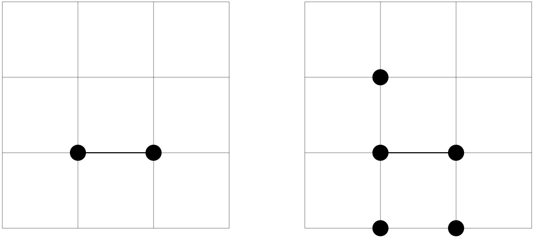


Figure: On the left is the Toric Graph. On the right are cross and face checks.

If both checks are unsatisfied, then flip the bit. Denote by κ the number of edges adjacent to those checks. Then the probability of failure is less than:

$$
\begin{aligned}
&(1-p)2^{-1} \cdot p^2 + p2^{-1}p \leq p^2 2 \\
&\leq p^{\frac{100}{99}}
\end{aligned}
$$

For p small enough.

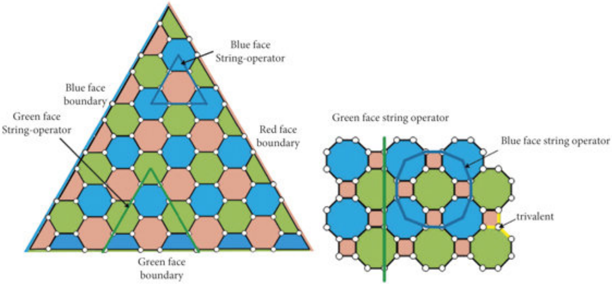
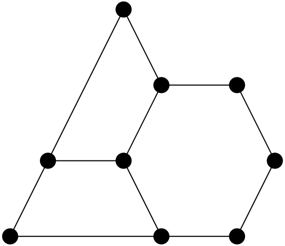


Figure: 2d color code



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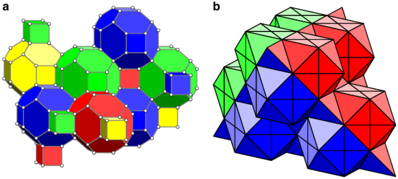

Figure 1 | Representations of the lattice geometry of the gauge color code. (**a**) In the primal picture qubits sit on the vertices of a four-valent lattice. The three-dimensional cells of the lattice are four colorable, that is, every cell can be assigned one of four colors such that it touches no other cell of the same color. (**b**) The gauge color code of linear size $L = 5$ drawn in the dual picture. Qubits lie on simplices of the lattice. The faces of the tetrahedra in the figure are given one of four colors such that no two faces of a given tetrahedron have the same color. The tetrahedra are stacked such that faces that touch always have the same color. For the gauge color code to encode a single logical qubit, the lattice must have four distinct, uniformly colored boundaries, as is shown in the figure. We give more details on the lattice construction in the dual picture in Supplementary Note 1 together with Supplementary Figs 1-6.

Figure: 3d color code